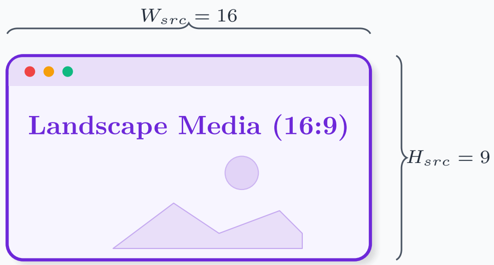
Aspect Ratio Inversion & Bounding Voids

Geometric derivations of Letterboxing and Pillarboxing in `object-fit: contain`

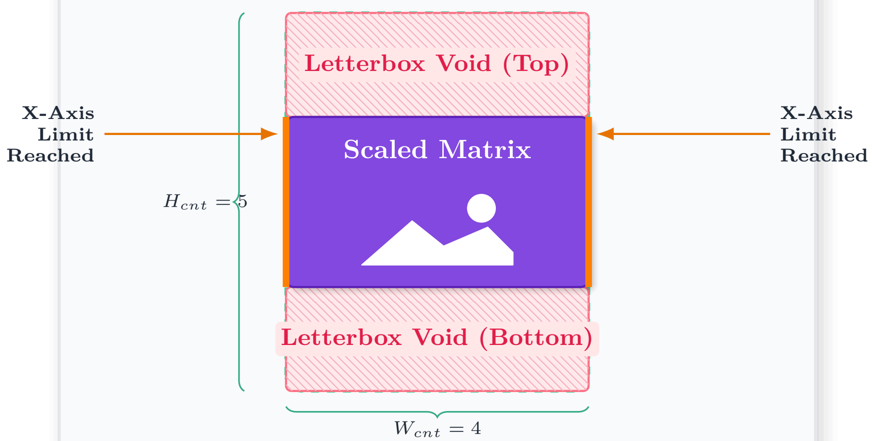
SCENARIO A: $R_{src} > R_{cnt}$



Media Aspect Ratio is **greater** than the Container. **Mathematical Output:** Width dictates the scale factor limit.

$$S_x = \frac{W_{cnt}}{W_{src}} = \frac{4}{16} = 0.25, \quad S_y = \frac{H_{cnt}}{H_{src}} = \frac{5}{9} \approx 0.556$$
$$\implies S = \min(S_x, S_y) = S_x$$

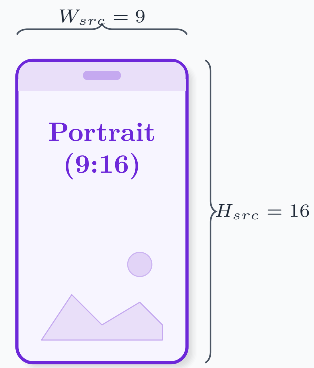
Portrait DOM Container (4:5)



The horizontal X-axis hits the container bounds first. To strictly maintain the 16:9 pixel aspect ratio, Y-axis interpolation stops early, leaving geometric voids above and below.

$\min(S_x, S_y)$ **Divergence**

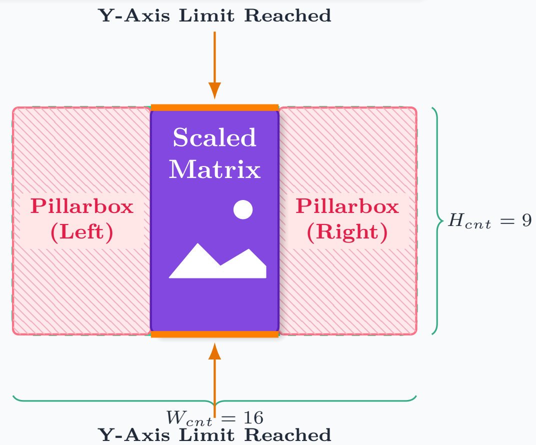
SCENARIO B: $R_{src} < R_{cnt}$



Media Aspect Ratio is **less** than the Container. **Mathematical Output:** Height dictates the scale factor limit.

$$S_x = \frac{W_{cnt}}{W_{src}} = \frac{16}{9} \approx 1.778, \quad S_y = \frac{H_{cnt}}{H_{src}} = \frac{9}{16} \approx 0.563$$
$$\implies S = \min(S_x, S_y) = S_y$$

Landscape DOM Container (16:9)



The vertical Y-axis hits the container bounds first. To strictly maintain the 9:16 pixel aspect ratio, X-axis interpolation stops early, generating geometric voids on the horizontal flanks.